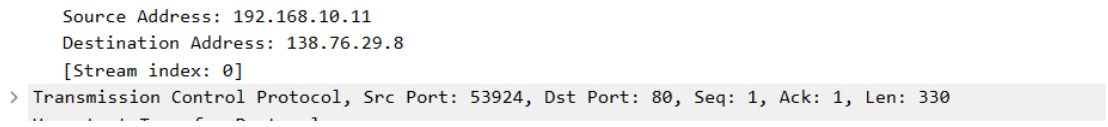
Assignment #7 - Wireshark NAT Lab

Part 1: NAT v9 Lab

Question #1: What is the IP address of the client that sends the HTTP GET request in the nat-inside-wireshark-trace1-1.pcapng trace? What is the source port number of the TCP segment in this datagram containing the HTTP GET request? What is the destination IP address of this HTTP GET request? What is the destination port number of the TCP segment in this datagram containing the HTTP GET request?

- IP Address of Client: 192.168.10.11
- Source Port #: 53924
- Destination IP Address: 138.76.29.8
- Destination Port Number: 80



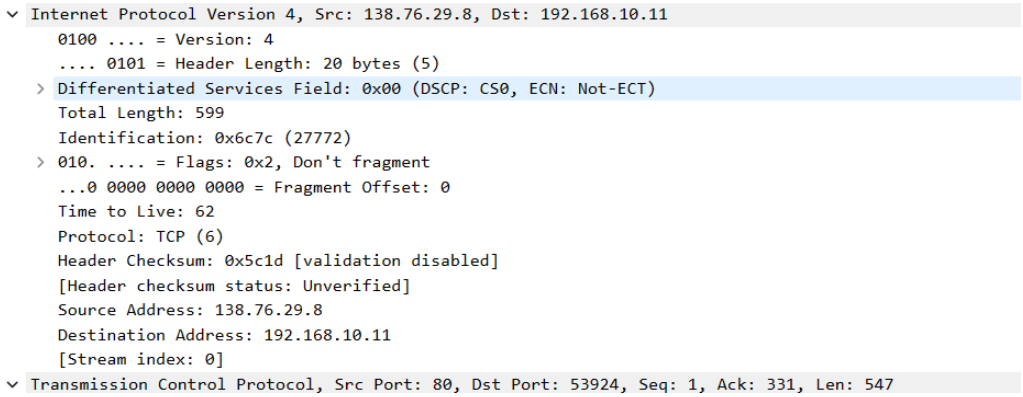
Question #2: At what time is the corresponding HTTP 200 OK message from the webserver forwarded by the NAT router to the client on the router's LAN side?

- Time: 23:50:27.774683377

Question #3: What are the source & destination IP addresses and TCP source and destination ports of the IP datagram carrying this HTTP 200 OK message?

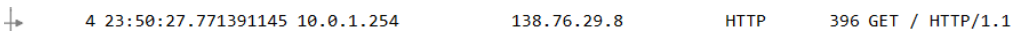
- Source IP: 138.76.29.8
- Destination IP: 192.168.10.11
- Source Port: 80

- Destination Port: 53924



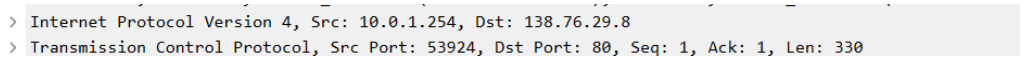
Question #4: At what time does this HTTP GET message appear in the nat-outside-wireshark-trace1-1.pcapng trace file?

- Time: 23:50:27.771391145



Question #5: What are the source and destination IP addresses and TCP source and destination port numbers on the IP datagram carrying this HTTP GET (as recorded in the nat-outside-wireshark-trace1-1.pcapng trace file)

- Source IP: 10.0.1.254
- Destination IP: 138.76.29.8
- Source Port: 53924
- Destination Port: 80



Question #6: Which of these four fields are different than in your answer to question 1 above?

- Source IP address field is different

Question #7: Are any fields in the HTTP GET message changed?

- Inside:

```
> Frame 4: Packet, 396 bytes on wire (3168 bits), 396 bytes captured (3168 bits) on interface eth1, id 0
> Ethernet II, Src: PCSSystemtec_89:c7:7c (08:00:27:89:c7:7c), Dst: PCSSystemtec_82:36:d7 (08:00:27:82:36
> Internet Protocol Version 4, Src: 192.168.10.11, Dst: 138.76.29.8
> Transmission Control Protocol, Src Port: 53924, Dst Port: 80, Seq: 1, Ack: 1, Len: 330
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: 138.76.29.8\r\n
    User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    \r\n
    [Response in frame: 6]
    [Full request URI: http://138.76.29.8/]
```

- Outside:

```
> Frame 4: Packet, 396 bytes on wire (3168 bits), 396 bytes captured (3168 bits) on interface eth1, id 0
> Ethernet II, Src: PCSSystemtec_89:c7:7c (08:00:27:89:c7:7c), Dst: PCSSystemtec_82:36:d7 (08:00:27:82:36
> Internet Protocol Version 4, Src: 192.168.10.11, Dst: 138.76.29.8
> Transmission Control Protocol, Src Port: 53924, Dst Port: 80, Seq: 1, Ack: 1, Len: 330
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: 138.76.29.8\r\n
    User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    \r\n
    [Response in frame: 6]
    [Full request URI: http://138.76.29.8/]
```

- It does not appear like any of the fields in the HTTP GET message has changed.

Question #8: Which of the following fields in the IP datagram carrying the HTTP GET are changed from the datagram received on the local area network (inside) to the corresponding datagram forwarded on the Internet side (outside) of the NAT router: Version, Header Length, Flags, Checksum?

- Inside:

```
Internet Protocol Version 4, Src: 192.168.10.11, Dst: 138.76.29.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 382
    Identification: 0x6296 (25238)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x64dc [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.10.11
    Destination Address: 138.76.29.8
    [Stream index: 0]
```

- Outside:

```
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 382
    Identification: 0x6296 (25238)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 63
    Protocol: TCP (6)
    Header Checksum: 0x2492 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.1.254
    Destination Address: 138.76.29.8
```

- Changes: Checksum has changed, the other 3 stayed the same

Question #9: At what time does this message appear in the nat-outside-wireshark-trace1-1.pcapng trace file? (The HTTP 200 OK)

- Time: 23:50:27.774660820

```
     6 23:50:27.774660820 138.76.29.8          10.0.1.254          HTTP      613 HTTP/1.1 200 OK  (text/html)
```

Question #10: What are the source and destination IP addresses and TCP source and destination port numbers on the IP datagram carrying this HTTP reply (”200 OK”) message (as recorded in the nat-outside-wireshark-trace1-1.pcapng trace file)?

- Source IP: 138.76.29.8
- Destination IP: 10.0.1.254
- Source Port: 80
- Destination Port: 53924

```
> Internet Protocol Version 4, Src: 138.76.29.8, Dst: 10.0.1.254
> Transmission Control Protocol, Src Port: 80, Dst Port: 53924, Seq: 1, Ack: 331, Len: 547
```

Question #11: What are the source and destination IP addresses and TCP source & destination port numbers on the IP datagram carrying the HTTP reply ("200 OK") that is forwarded from the router to the destination host in the right of Figure 1?

- Source IP: 138.76.29.8
- Destination IP: 192.168.10.11
- Source Port: 80
- Destination Port: 53924

Part 2: Assume we have an organization with 450 users. These users are from three divisions, namely Engineering (EN), Human Resources (HR), and Sales (SA). While each of EN and HR have 128 users, SA has 194 users. Assume that an ISP has a large block of addresses (192.168.184.0/21) and it needs to assign a block of addresses to this organization using classless addressing. (9 points)

a. Determine the block of addresses which is allocated to this organization by the ISP, including the first and last addresses as well as the number of required addresses. (2 points)

- Block: 192.168.184.0/23
- First Address: 192.168.184.0
- Last Address: 192.168.185.255
- Required Address Amount: 512 addresses
  - $2^8 = 256, 2^9 = 512$
  - 450 is not a power of 2, so we go to the next highest, which would be 512

b. How many addresses from the large block of addresses (i.e., the original block of addresses) are unused? (1 point)

- Unused Addresses: 1536 addresses
- $2^{11} - 2^9 = 1536$

c. Assume you are splitting the allocated block of addresses to the organization into three subnetworks, each of which corresponds to a division. Indicate the first and last addresses as well as the number of required addresses for each division. (6 points)

- EN → 128 total people
  - Requires: 128 addresses
  - First: 192.168.185.128
  - Last: 192.168.185.255
  - Block: 192.168.185.128/25
- HR → 128
  - Requires: 128 addresses
  - First: 192.168.185.0
  - Last: 192.168.185.127
  - Block: 192.168.185.0/25
- SA → 194
  - Requires: 256 addresses
  - First: 192.168.184.0
  - Last: 192.168.184.255
  - Block: 192.168.184.0/24

NOTE: For simplicity, assume all of the addresses in a block are usable, including the broadcast and network addresses. For example, if a department requires 16 addresses, you do not need to consider adding two extra addresses for broadcast and network address.